

Project Links and Details

1. **Github:** <https://github.com/Mohammad-Shehnaj/AIDD-Project>

2. **Video Presentation:** https://studentncirl-my.sharepoint.com/:v/g/personal/x23305762_student_ncirl_ie/EQF5BMYsok5DiP2S-IbowqQBHA0yrknLDkvQSaN-IgurMw?e=2dKTpf

3. **Work Distribution:**

We (*Mohammad Shehnaj & Shayshank Rathore*) collaborated throughout the project with shared ownership and mutual support across all major tasks.

Both of us **equally contributed** to writing the report, analyzing results and building the final presentation. Regular discussions ensured that all approaches were well-aligned and consistent. Here's a breakdown of our primary contributions:

➤ **Mohammad Shehnaj**

- Led the development of the AMPL standalone model, including mathematical formulation, constraint setting and solver configuration using CPLEX.
- Designed and executed the Python + AMPL hybrid integration, including automated .dat file generation and subprocess execution.
- Contributed significantly to report structuring, writing, diagram design and overall formatting.

➤ **Shayshank Rathore**

- Led the Python-based Genetic Algorithm implementation including data preprocessing, population design, fitness evaluation and performance analysis.
- Responsible for quarterly evaluation, return visualization and cumulative performance comparison.
- Assisted in hybrid integration and collaborated on report writing and slide preparation.

Task Responsibility Table:

Task	Mohammad Shehnaj	Shayshank Rathore
Data Preprocessing	☒	☒
Genetic Algorithm Implementation	☒	☒ (Lead)
AMPL Model & Optimization	☒ (Lead)	☒
Python + AMPL Hybrid Integration	☒	☒
Report Writing	☒	☒
Performance Evaluation	☒	☒
Final Review & QA	☒	☒
Presentation Design	☒	☒